When the delta is fake

A construction for catching no-op “wins” in agent evaluation

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ABSTRACT

Agent benchmarks increasingly score autonomous repair and tool-use by final state: instruction, environment, tests, oracle (Merrill et al., 2026). That unit is necessary. It is not sufficient. A run can satisfy a schema, print a large “delta,” or turn tests green while the underlying method is a no-op, the fixture has been edited, or the reported control was never recomputed.

This note proposes a small integrity layer for agent-task authorship (a pass-bar) and a concrete construction that makes fake improvement hard to hide. For ranking AUROC under a linear scorer, pure mean shift need not change order. Mean-only adaptation can look serious with delta still zero. Pairing that invariance with a train-only label shortcut that flips at evaluation, plus a frozen verifier-recomputed control, forces any passing method to drop the shortcut rather than rewrite the report.

Three public tasks instantiate the pass-bar. One staged transcript uses an explicit process rubric. Methodological claim. Synthetic fixtures. No frontier leaderboard.

Claim in one line. A green-looking delta can be fake. Ranking invariance + shortcut flip + frozen recomputed control makes the fake fail.

1. The problem

Outcome harnesses answer: did the final state match the tests? They do not answer: was the pass earned by solving the intended difficulty, or by negotiating the measurement?

1. **Schema pass, method nop.** Artifacts match a JSON shape while control and model fields are identical.
2. **Cosmetic adaptation.** Normalization that cannot change the scored ranking metric.
3. **Control capture.** The agent weakens the baseline instead of beating a frozen one.
4. **Fixture negotiation.** Editing goldens so tests agree with a broken implementation.
5. **First-red fixation.** Repair stops at the first failing assertion.

Benchmarks that only publish pass rate under these modes overstate progress. The contribution is a design response, not a new 100-task corpus.

2. Proposed pass-bar

Criterion	Meaning
Nop failure	Starter fails for a failure mode stateable in one sentence
Oracle success	Reference solution clears every test

Criterion	Meaning
Artifact-backed score	Verifier recomputes control or expected state
Calibration record	Documented starter modes and blocked shortcuts

Relative to Terminal-Bench's unit, this is an authorship integrity layer. It makes a class of false greens expensive.

3. The construction: fake deltas under ranking metrics

3.1 Ranking invariance

Adding a constant to every score leaves pairwise order unchanged, hence leaves AUROC unchanged. A covariate mean shift absorbed into the bias behaves the same way for ranking. Mean-only "adaptation" is therefore not evidence of distribution-shift competence when the gate is AUROC delta.

3.2 Shortcut flip + frozen control

1. Plant a strong label association in a feature on training sites.
2. Flip that association on the evaluation site.
3. Freeze a control on all raw features and recompute it in the verifier.
4. Require model – control above a calibrated threshold with tight control match.

Z-score-only methods fail. Methods that drop the shortcut can clear a large delta. Editing the report cannot lower the control.

3.3 Checked-in instance (site-shift-meter)

Quantity	Value
`control_auroc`	0.0108
`model_auroc`	0.9766
`delta_auroc`	0.9658
Gate	`>= 0.08`, control match `1e-6`
`n_eval`	180

4. Instantiations

The construction is metric-specific. The pass-bar is not. Further tasks show the same integrity pattern: event replay with read-only fixtures and recomputed state; dual-consumer ring where single-consumer greens do not imply the contract; a staged fail transcript with signed process rubric (positive mass 13).

5. Relation to existing stacks

Terminal-Bench provides the outcome unit. Harbor executes. Inspect composes dataset / agent / scorer. This note adds authorship constraints those stacks leave open.

6. Limitations

- One primary construction (ranking AUROC + shortcut flip).

- Three illustrative tasks, synthetic data, no multi-model sweep.
- Staged transcript. Gate calibrated to these CSVs.
- No capability or safety claims about models.

7. Reproduction

```
git clone https://github.com/gyanu2507/agent-eval-suite
cd agent-eval-suite
python3 -m pytest -q -p no:randomly tasks/*/tests
python3 tasks/site-shift-meter/solution/train_and_eval.py
```

References

Merrill, M. A., Shaw, A. G., Carlini, N., et al. (2026). *Terminal-Bench*. arXiv:2601.11868. https://doi.org/10.48550/arXiv.2601.11868

Harbor Framework. https://github.com/harbor-framework/harbor

UK AI Security Institute. *Inspect*. https://inspect.aisi.org.uk/

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